O-M-232d

JUNE 16, 1960

SUPERSEDING
Int. Fed. Spec. O–M–00232c (ARMY–CE)
May 19, 1959 and
Fed. Spec. O–M–232b
July 22, 1958

FEDERAL SPECIFICATION

METHANOL (METHYL ALCOHOL)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers methanol of a technical grade and synthetic grades.

1.2 Classification.—Methanol shall be of the following grades, as specified (see 6.2):

Grade A. — Synthetic, 99.85 percent by weight (solvent use).
Grade AA. — Synthetic, 99.85 percent by weight (hydrogen-carbon dioxide generation use).
Grade B. — Technical, 99.0 percent by weight (solvent use).
Grade C. — Wood alcohol (denaturing grade).

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

UU–P–236—Paper, Filter, Analytical.
PPP–B–636—Box, Fiberboard.
PPP–C–96—Cans, Metal, 28-Gage and Lighter.
PPP–D–729 — Drums, Metal, 55-Gallon (for Shipment of Noncorrosive Materials).
PPP–D–760—Drums and Pails, Metal (5 and 16.64 Gallons).

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 791—Lubricants, Liquid Fuels, and Related Products; Methods of Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

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Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

MIL–STD–147 — Palletized Unit Loads (40" x 48" 4-Way Partial and 4-Way Pallets).

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.—The following publications form a part of this specification. Unless otherwise indicated, the issues in effect on date of invitation for bids shall apply.

Interstate Commerce Commission,

49–CFR–71–78 — Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(Now a part of the Code of Federal Regulations (1957 Edition) copies are available from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders should cite "49 CFR 71–78".)

U. S. Treasury Department,

Internal Revenue Service Regulations, 26–CFR–212 — Formulas for Denatured Alcohol.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. Orders should cite "26 CFR 212".)

American Society for Testing Materials,

ASTM Designation: D 1078—Distillation Range of Lacquer Solvents and Diluents.

ASTM Designation: D1209 — Color of Clear Liquids (Platinum-Cobalt Scale).

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pa.)

Uniform Freight Classification Rules.

(Application for copies should be addressed to Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N. Y.)

National Motor Freight Classification Rules.

(Application for copies should be addresse American Trucking Associations, Inc., 1424 Street, N. W., Washington, D. C.)

3. REQUIREMENTS

3.1 Grades A and B. — Grades A an methanol shall be as specified in table I

3.2 Grade AA.—Grade AA methanol s be as specified for grade A, except that alcohol shall contain not more than 0 percent ethanol and not more than 0 percent acetone.

3.3 Grade C.—Grade C wood alcohol naturing grade) shall comply with the quirements of SDA1 and SDA2A of 26–C 212, Formulas for Denatured Alcohols.

4. SAMPLING, INSPECTION, AND T PROCEDURES

4.1 Unless otherwise specified herein, supplier is responsible for the perform of all inspection requirements prior to mission for Government inspection and ceptance. Except as otherwise specified, supplier may utilize his own facilities or commercial laboratory acceptable to the ernment. Inspection records of the exan tions and tests shall be kept complete available to the Government as specifie the contract or order.

4.2 Classification of inspection.—Inspe shall be classified as follows:

(a) Acceptance inspection of methan

(b) Acceptance inspection of prepar for delivery.

4.3 Acceptance inspection.

4.3.1 *Sampling.*

4.3.1.1 *For examination.*—Sampling fo amination shall be in accordance with tary Standard MIL–STD–105.

4.3.1.2 *For tests.*

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TABLE I.—*Requirements for Grade A and Grade B*

Characteristic	Grade A		Grade B	
	Requirement	Test paragraph	Requirement	Test paragraph
Acetone and aldehydes	Not more than 0.003 percent	4.3.3.2	----------	-----
Acidity	Not more than 0.003 percent as acetic acid	4.3.3.4.1	Neutral to blue litmus	4.3.3.4.3
Alkalinity	Not more than 0.003 percent as NH_3	4.3.3.4.2	Neutral to red litmus	4.3.3.4.3
Appearance	Clear, free from suspended matter or sediment	4.3.3.5	Clear, free from suspended matter or sediment	4.3.3.5
Carbonizable substances	Not darker than 0.000125 N iodine solution when tested with sulfuric acid	4.3.3.6	----------	-----
Color	Not darker than 5 units on the platinum-cobalt (APHA) scale	4.3.3.7.1	Not darker than solution containing 0.0030 gram potassium dichromate in 1 liter distilled water	4.3.3.7.2
Distillation range	Not more than 1° C. and shall include 64.6° C. ± 0.10° C. at 760 mm. Hg	4.3.3.8	Not more than 2° C. and shall include 64.7° C. at 760 mm. Hg	4.3.3.8
Hydrocarbons	To pass test	4.3.3.9	----------	-----
Specific gravity (maximum)	0.7932 at 20°/20° C.	4.3.3.10	0.7953 at 20°/20° C.	4.3.3.10
Percent methanol by weight	99.85	4.3.3.10	99.0	4.3.3.10
Nonvolatile content, (maximum)	0.0010 percent	4.3.3.11.1	Not more than 0.002 gm. from a 100-ml. sample	4.3.3.11.2
Odor	Characteristic and free from foreign odors	4.3.3.12.1	Characteristic of methanol free from objectionable odors, mild empyreumatic odor not objectionable. No residual odor after evaporation from 100-ml. sample	4.3.3.12.2
Permanganate test	No discharge of pink color in 30 minutes	4.3.3.13 to 4.3.3.13.3 inclusive	----------	-----
Spot test	----------	-----	No spotting or staining	4.3.3.14

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4.3.1.2.1 *Drums and small containers.* — From each lot, two containers shall be selected at random. From each of the two containers, one-quart specimens shall be taken and placed in separate, clean, dry, glass containers.

4.3.1.2.2 *Tank cars.* — Samples shall be taken in accordance with method 8001, Federal Test Method Standard No. 791.

4.3.2 *Examination.*—Samples selected in accordance with 4.3.1.1 shall be examined for the following defects. Lot acceptance or rejection shall be based upon an AQL of 2.5 percent defective.

Defects

1. Unsatisfactory markings.
2. Incomplete fill.

4.3.3 *Tests.*—Samples selected as specified in 4.3.1.2 shall be tested as specified hereinafter. Visual comparisons shall be made using a white light source. If any specimen fails any test, the lot which the specimen represents shall be rejected.

4.3.3.1 *Chemical reagents.* — Wherever chemical reagents are specified in the following tests, analytical reagent grade shall be used.

4.3.3.2 *Acetone and aldehydes.*

4.3.3.2.1 *Nessler's reagent.*—Stock solution A shall be made by dissolving 270 grams of sodium hydroxide in distilled water and diluting to 1 liter with distilled water. Stock solution B shall be prepared by dissolving 36 grams of potassium iodide and 13.6 grams of mercuric chloride in distilled water and diluting to 500 milliliters with distilled water. Prior to use, three parts of solution A shall be mixed with five parts of solution B to make the Nessler's reagent. It shall be allowed to stand, preferably over night, until the solution is clear.

4.3.3.2.2 *Procedure.* — Five milliliters of Nessler's reagent shall be added to a solution of 1 milliliter of specimen in 4 milliliters of distilled water. If turbidity is noted, *it* be compared to the standard specified he Comparative standard acetone solution be made up as follows: Dilute 5.0 milli of c. p. acetone with water to make 1.0 mix thoroughly — this is solution A. 1.0 milliliter of solution A and make up water to 655 milliliters; mix thoroughl this is solution B. Use 5 milliliters of tion B for acetone standard which sh contain 0.08 milligram of acetone. If the bidity is greater than that produced by milligram of acetone in 5 milliliters of w when 5 milliliters of Nessler's reager added, the specimen shall have failed test.

4.3.3.3 *Acetone and ethanol (grade A* —Acetone and ethanol shall be determ by the elution method of gas chromatogr using internal standards.

4.3.3.3.1 *Apparatus and operating c tions.* — Any conventional chromatogra technique utilizing an instrument at equal in performance to the Perkin E Vapor Phase Fractometer Model 154 sha used. A two meter elution column using t ethylene glycol dimethyl ether (TEDME equal as the liquid phase shall be used u the following operating conditions:

Temperature	plus 5
Pressure	Adjust to produce helium rate of 80 plus or minus 10
Sample size	0.2
Highest sensitivity range	

4.3.3.3.2 *Qualitative analysis.*—Qualit detection of ethanol and acetone may be n by adding 0.001 percent ethanol and ace to a specimen of the unknown and compa the position of these curves to those in unknown specimen under identical opera conditions. If ethanol or acetone is pre in the unknown, the curves will fall in actly the same position relative to the m anol curve as those in the standard in o to have an acceptable product. If the cu fall outside this limit, the specimen s have failed the test.

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4.3.3.3.3 *Quantitative analysis.*

(a) Internal standards for quantitative analysis curves shall be run on standard specimens prepared by adding pure ethanol or acetone in 0.001 percent by volume increments. The area under the peak is equivalent to the concentration of ethanol or acetone in the specimen, and the peak height is proportional to the concentration. Using a simple ratio, the concentration of ethanol or acetone can be calculated.

(b) The area under the ethanol or acetone curve of the unknown shall be no more than one half that under the ethanol and acetone curve of the 0.001 percent ethanol or methanol standard specimen. If the results of the comparison of calculations differ from the value shown in table I, the specimen shall have failed the test.

4.3.3.4 *Acidity or alkalinity.*

4.3.3.4.1 *Acidity (grade A).*—To 25 milliliters of water and 10 milliliters of specimen, add 0.5 milliliter of 1-percent alcohol phenolphthalein indicator solution and 0.01 sodium hydroxide (NaOH) until a slight pink color persists after the mixture is shaken for ½ minute. Add 25 milliliters of specimen, mix well, then titrate with 0.01 *N* NaOH until pink color is restored.

$$\text{Percent acidity as acetic acid} = \frac{(\text{Milliliters of NaOH}) \times (\text{Normality of NaOH}) \times 6.005}{(\text{Milliliters of specimen}) \times 0.79}$$

If the results of comparison of calculations differ from the value shown in table I, the specimen shall have failed the test.

4.3.3.4.2 *Alkalinity (grades A and AA).*—To 25 milliliters of water, add 1 drop of 0.1 percent methyl red indicator solution (60 percent alcohol) and 0.01 *N* acid or base until the solution is just pink. Add 25 milliliters of specimen, mix well, then titrate with 0.01 *N* sulfuric acid until the pink color is restored. Calculate the percent NH_3 by weight.

$$\text{Percent alkalinity as } NH_3 = \frac{(\text{Milliliters of } H_2SO_4) \times (\text{Normality of } H_2SO_4) \times 1.703}{(\text{Milliliters of specimen}) \times 0.79}$$

If the results of comparison of calculations differ from the value shown in table I, the specimen shall have failed the test.

4.3.3.4.3 *Acidity-alkalinity (grade B).*—Mix 10 milliliters of sample with an equal volume of freshly boiled and cooled distilled water. Test immediately with red and blue litmus paper. If change in color of the litmus paper is noted the specimen shall have failed the test.

4.3.3.5 *Appearance.*—The specimen shall be thoroughly mixed and shall be examined by transmitted light for color, clarity, i.e., suspended matter or sediment. This is most readily done by pouring some of the specimen into a test tube or Nessler tube. If the specimen is off color, not clear, or contains suspended matter or sediment the specimen shall have failed the test.

4.3.3.6 *Carbonizable substances (grades A and AA).*—A tube shall be drawn down to a small orifice which will deliver 5 milliliters of concentrated sulfuric acid in 5 minutes. With this tube, 5 milliliters of concentrated sulfuric acid shall be added to 5 milliliters of sample in a long test tube. The resulting solution shall be transferred to a 1-ounce, glass-stoppered, square, clear glass bottle approximately 3 centimeters by 3 centimeters by 6 centimeters high. Color comparison shall be made with a 0.000125 *N* solution of iodine in a similar glass bottle. The comparison solution shall be made by dissolving 12.7 grams of iodine and 10.9 grams of potassium iodide in a liter of distilled water, and then diluting 5 milliliters of this 0.1 *N* solution to 4,000 milliliters with distilled water. If the color is not visually equivalent upon the comparison of the standard and specimen the specimen shall have failed the test.

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4.3.3.7 *Color.*

4.3.3.7.1 *Grades A and AA.*—Determine the color in accordance with ASTM method D 1209.

4.3.3.7.2 *Grade B.*—Prepare a standard solution by dissolving 0.003 gram of potassium dichromate in a liter of distilled water at room temperature (20° to 30° C.). Compare the color of the methanol with that of the standard solution at room temperature (20° to 30° C.) in 50-milliliter Nessler tubes.

4.3.3.7.3 *Results.*—If the color is not visually equivalent upon the comparison of the standard and specimen, the specimen shall have failed the test.

4.3.3.8 *Distillation range.*—Determine the distillation range in accordance with ASTM method D–1078, with the following exceptions:

(a) A gas burner is preferable; an electric heater may cause high results due to superheating. It is recommended that boiling stone or glass beads be used to prevent bumping if an electric heater is used.

(b) Measure the 100 milliliters of specimen and distillate at room temperature (without water bath).

(c) Determine temperature using JL–4 type thermometer having a range of 45°-75° C. in increments of 0.1° C.

If the distillation range differs from the value shown in table I, the specimen shall have failed the test.

4.3.3.9 *Hydrocarbons (grades A and AA).*—Dilute 15 milliliters of the specimen to 45 milliliters with distilled water in a Nessler tube. If cloudiness or opalescence is observed at any time during 20 minutes of standing, the specimen shall have failed the test.

4.3.3.10 *Specific gravity (methanol content).*—Determine the specific gravity with a pycnometer of about 50-milliliter capacity or by any other method accurate to the fourth decimal place. In the temperature rang to 25° C., the specific gravity changes 0. per degree centigrade. Determine the anol content from the specific gravity specimen. If the methanol content c from the value shown in table I, the spe shall have failed the test.

4.3.3.11 *Nonvolatile content.*

4.3.3.11.1 *Grades A and AA.*—Five dred milliliters of the specimen shall be orated in a weighed platinum or other able evaporating dish on a steam bath evaporating dish shall then be heated oven at 105° to 110° C. for ½ hour, in a desiccator, and reweighed.

4.3.3.11.2 *Grade B.*—Evaporate 100 liters of the thoroughly mixed specim a tared porcelain evaporating dish to dr on the steam bath. Heat for 30 minut an oven at 105° to 110° C., cool, weigh residue shall weigh not more than gram.

4.3.3.11.3 *Rejection criteria.*—If the ount of nonvolatile matter is greater specified in table I, the specimen shall failed the test

4.3.3.12 *Odor.*

4.3.3.12.1 *Grades A and AA.*—Place 5 liters of specimen in a 1- by 4-inch tube, insert stopper, and shake vigor for a few seconds; then let stand fo minutes. Open and cautiously examin odor of foreign materials.

4.3.3.12.2 *Grade B.*—Note the odor o material in the container. Dip uniform s of medium-texture, double-acid-washed, titative filter paper (similar to type II, 6, of Federal Specification UU–P–236) i sample. Allow to dry in a well-venti room at 20° to 30° C. for 2 hours. Exa the strips at the beginning of the dr period and at 15-minute intervals durin drying period. At the end of the dryin riod, the specimen shall be examined fo sidual odor.

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4.3.3.12.3 *Rejection criteria.*—If the specimen has an odor of foreign matter, the specimen shall have failed the test.

4.3.3.13 *Permanganate test (grades A and AA).*

4.3.3.13.1 *Apparatus.*—A constant-temperature bath of suitable size to accommodate a 50-milliliter, tall form, glass-stoppered cylinder shall be provided. It shall be adjusted to 15.5° ± 0.5° C. The 50-milliliter cylinders shall be cleaned with 15 to 20 milliliters of concentrated hydrochloric acid, rinsed six times with tap water, rinsed twice with distilled water, and finally rinsed with some of the specimen to be tested.

4.3.3.13.2 *Reagent.*—Permanganate reagent shall be prepared by dissolving 0.2 gram of potassium permanganate in 1,000 milliliters of freshly boiled distilled water. This shall be protected from light and shall be renewed at least every 3 days.

4.3.3.13.3 *Procedure.* — A 50-milliliter, glass-stoppered cylinder shall be filled beyond the mark with the specimen to be tested and placed in the constant temperature bath. The water level in the bath shall be approximately 1 inch below the neck of the glass-stoppered cylinder. When the specimen has reached 15.5° ± 0.5° C., the level shall be brought to the 50-milliliter mark. Two milliliters of the permanganate reagent shall be added from a pipette. The cylinder shall be glass-stoppered, inverted once to mix the contents, and returned to the bath. The time from addition of the permanganate until discharge of the pink color shall be determined, if less than the limit of 30 minutes. The cylinder shall be protected from light during this time. The following may be used as a blank: To 5 milliliters of cobaltous chloride (50 grams per liter), add 7 milliliters of uranium nitrate (40 grams per liter); then dilute with distilled water to 50 milliliters.

4.3.3.13.4 *Rejection criteria.*—If the color of the specimen and the standard differ upon visual comparison, the specimen shall have failed the test.

4.3.3.14 *Spot test (grade B).*—Support a piece of medium texture, double-acid-washed, quantitative filter paper (similar to type II, class 6 of Fed. Spec. UU–P–236) so that the center is free from contact with any undersurface. Place five drops of specimen in the center of the filter paper. Allow to evaporate for 2 hours in a well-ventilated room at 20° C. to 30° C. If there is spotting or staining of the paper at the end of the drying period, the specimen shall have failed the test.

4.4 Acceptance inspection and tests of preparation for delivery.—The packaging, packing, and marking shall be examined and tested for compliance with section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 For civil agency procurement the definitions and applications of the levels of packaging and packing shall be in accordance with Federal Standard No. 102.

5.2 Packaging.—Packaging shall be level A or C, as specified (see 6.2).

5.2.1 *Level A.*—The methanol shall be packaged in one-quart or one-gallon cans conforming to Federal Specification PPP–C–96, type V, class 4, oblong shape, or in 5-gallon drums conforming to Federal Specification PPP–D–760, type I, class 1 or 2, with screw cap closure, or in 55-gallon drums conforming to Federal Specification PPP–D–729, type II, as specified (see 6.2). Unless otherwise specified (see 6.2), plan B coating with side seam stripping for one-quart cans and one-gallon cans shall be required in accordance with Federal Specification PPP–C–96. One-gallon cans shall be provided with a formed bridge type-handle securely affixed to the top. The handle shall be protectively coated to resist corrosion. The color of the exterior coating on the 5-gallon drums shall be as specified (see 6.2). The wire handles or bails of the 5-gallon drums shall be galvanized or protectively coated to resist corrosion.

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5.2.2 *Level* C.—Methanol shall be packaged in containers conforming to the supplier's standard practice of the size specified in 5.1.1 and when specified (see 6.2), in bulk quantity in railroad tank cars.

5.3 Packing.—Packing shall be level A, B, or C, as specified (see 6.2).

5.3.1 *Level A.*—Methanol in one-quart and one-gallon cans shall be packed in accordance with the appendix of Federal Specification PPP–C–96 as specified for overseas shipment. When specified (see 6.2), six 1-gallon cans of methanol shall be packed in an exterior shipping container conforming to Federal Specification PPP–B–636. Five-gallon drums and 55-gallon drums shall not require additional packing. When specified (see 6.2), drums shall be palletized in accordance with Military Standard MIL–STD–147.

5.3.2 *Level B.*—Methanol in one-quart and one-gallon cans shall be packed in accordance with the appendix of Federal Specification PPP–C–96 as specified for domestic shipment. When specified (see 6.2), six 1-gallon cans of methanol shall be packed in an exterior container conforming to Federal Specification PPP–B–636. Five-gallon drums and 55-gallon drums shall not require additional packing. When specified (see 6.2), drums shall be palletized in accordance with Military Standard MIL–STD–147.

5.3.3 *Level C.*—Methanol in one-quart and one-gallon cans shall be packed in containers complying with Uniform Freight Classification Rules or with National Motor Freight Classification Rules or with other carrier rules applicable to the mode of transportation. Five-gallon drums and 55-gallon drums shall not require additional packing.

5.4 Marking.—Each container and palletized load shall be labeled as shown in figure 1 and marked in accordance with 49 CFR–71 – 78, inclusive. The label shall be waterproofed in accordance with Military Standard MIL–STD–129.

5.4.1 *Civil marking.*—In addition marking specified in 5.4, each containe be marked in accordance with Federal ard No. 123.

5.4.2 *Military marking.*—In addition special marking required by the contr order and as specified in 5.4, each con and palletized load shall be marked in a ance with Military Standard MIL–ST

6. NOTES

6.1 Intended use.

6.1.1 *Grades A and B.*—Grades A methanol are intended to be used as trial solvents.

6.1.2 *Grade AA.*—Grade AA meth intended to be used for the generat hydrogen and carbon dioxide in a ge conforming to Military Specification G–10157.

6.1.3 *Grade C.*—Grade C wood alc a product of destructive distillation o and is intended to be used as a dena which is a wood distillate conforming ternal Revenue Service Regulations 26 212.

6.2 Ordering data. — Procuremen should specify the following:

(*a*) Title, number, and date of this fication.
(*b*) Grade required (see 1.2).
(*c*) Administrative provisions for tion records (see 4.1).
(*d*) Level of packaging and level o ing required (see 5.2 and 5
(*e*) Size of container required (see
(*f*) When plan B coating and sid stripping is not required (see
(*g*) Color of exterior coating of 5 drums (see 5.2.1).
(*h*) When bulk shipment is requir 5.2.2).

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(*i*) Shipping containers required (see 5.3.1 and 5.3.2).

(*j*) When palletized loads are required (see 5.3.1 and 5.3.2).

6.3 Basis of purchase. — The methanol should be purchased by volume, the unit being the amount of fluid at plus 60° F. occupying 231 cubic inches.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded, by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTEREST:

Navy—Sh.

DANGER! FLAMMABLE
VAPOR HARMFUL
MAY BE FATAL OR CAUSE BLINDNESS, IF SWALLOWED
CANNOT BE MADE NONPOISONOUS.

Keep away from heat and open flame.
Keep container closed.
Avoid prolonged or repeated breathing of vapor.

Use only with adequate ventilation.



POISON



First Aid — Antidote

If swallowed: Give a tablespoonful of salt in a glass of warm water and repeat until vomit fluid is clear. Give two teaspoonfuls of baking soda in a glass of water. Have patient lie down and keep warm. Cover eyes to exclude light. CALL A PHYSICIAN.

FIGURE 1.—Poison label for methanol

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